Ash L. Gilbert

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Education

Ph.D. (Atmospheric and Oceanic Sciences) at **University of Colorado-Boulder** 2022 – 2027
M.S. (Atmospheric and Oceanic Sciences) at **University of Colorado-Boulder** 2022 – 2025
B.S.E., *summa cum laude* (Climate and Meteorology, conc. Meteorology) at **University of Michigan** 2018 – 2022
Minors in Computer Science and Latin

Research Interests

- polar cloud processes
- Community Earth System Model (CESM)
- Arctic climate
- wind nudging
- climate model development
- internal climate variability
- climate change attribution
- atmosphere-sea ice interactions
- wildfire-climate interactions

Research Experience

Graduate Research Assistant, University of Colorado-Boulder 2022 – Present
Advisor: Professor Jennifer Kay

Project 1 (2023 – 2025): **A novel model hierarchy isolates the limited effect of supercooled liquid cloud optics on infrared radiation**

- Collaborator: Dr. Penny Rowe
- Evaluated the impact of supercooled liquid water optical properties on the Arctic climate via a hierarchy of climate models (CESM), ordered by level of dynamical constraint
- Identified a substantial effect from the optics in constrained dynamical models, but not in dynamically unconstrained models
- Proposed this model hierarchy as a method to assess the effect of model physics changes on a model climate
- First-author manuscript of this project has been published

Project 2 (2024 – 2025): **Observed winds alone cannot explain recent Arctic warming and sea ice loss**

- Collaborators: Professor Edward Blanchard-Wrigglesworth, Dr. Marika Holland, Dr. David Bailey, Professor Alexandra Jahn, and Dr. David Schneider
- Assessed the contribution of winds to recent Arctic warming and sea ice loss using wind nudging in CESM and comparing to observations and the pre-industrial CESM control
- Evaluated the change in wind contribution for an increase and decrease in mean state sea ice thickness
- Developed a tuned version of CESM2 that has a pre-industrial climate and modern-day mean state sea ice
- First-author manuscript of this project has been published

Project 3 (2025 – Present): **Internal and anthropogenic contributions to recent (1950-2024) Arctic temperature and sea ice change**

- Collaborators: Professor Edward Blanchard-Wrigglesworth
- Assessed the contribution of winds, greenhouse gases, and aerosols to recent Arctic warming and sea ice loss using wind nudging in CESM and single-forcing attribution framework
- Evaluated the performance of new wind nudging runs against observations and large ensemble runs
- First-author manuscript of this project was submitted in summer 2026

Project 4 (2026 – Present): **Precipitation detection using EarthCARE**

- Collaborators: Professor Hélène Chepfer
- Visited Laboratoire de Météorologie Dynamique at École Polytechnique in Palasieau, France for a month during summer 2026 to begin collaboration
- Analyzing precipitation from EarthCARE Cloud Profiling Radar to understand additional information gained from EarthCARE's increased vertical resolution
- Project is still being scoped, but will likely focus on the Arctic

Project 5 (2026 – Present): **Theoretical basis of wind nudging**

- Collaborators: Professor Ian Eisenmann and Professor William Chapman
- Theoretical modeling project designed to probe how wind nudging affects a coupled climate model

Research Assistant, University of Michigan 2021 – 2022
Project: **UFS-FVCOM Lake Effect Snow Case Study: February 5-7, 2021**
Advisor: Professor Christiane Jablonowski

- Analyzed a case study of lake effect snow forecasting for the UFS-SRW model coupled to the FV3COM, a lake dynamics model
- Ran various configurations of the coupled UFS-FV3COM model on the NCAR Cheyenne supercomputer
- Compared model results to reanalysis data and snow accumulation observations to evaluate model skill
- Contributed to manuscript that included this case study, which is in review

NSF Oceanic Sciences (Virtual) REU Student, NorthWest Research Associates 2020
Project: **Influence of Temperature Dependence of Cloud Optical Properties on Simulated IR Flux at South Pole, Antarctica**
Advisor: Dr. Penny Rowe

- Worked under a mentor to find the influence of temperature dependence of cloud optical properties on simulated IR flux
- Received anti-discrimination, anti-harassment, DEI, and research ethics training through a series of professional development seminars provided by NSF
- Gave research talk as final part of the REU
- Presented poster on work in 2021 at national conference for American Meteorological Society

Research Assistant, University of Michigan 2019 – 2023
Project: **Observing and Modeling Drainage Networks from Supraglacial Lakes on Russell Glacier, West Greenland**
Advisor: Professor Jeremy Bassis

- Determined the controls on supraglacial lake formation through observational analysis and modeling to further knowledge of ice sheet ablation
- Identified the observed spatio-temporal patterns in supraglacial lake formation and air temperature with ArcGIS
- Modeled how supraglacial lakes drain to replicate observed patterns using Python software and ArcGIS
- Presented poster on work in 2020 and 2021 at national conference for American Geophysical Union
- Participated in weekly lab meetings discussing relevant research papers and professional development topics
- Wrote and submitted a research paper for publication from this project (rejected)

Expedition to Greenland, University of Michigan 2019
Advisors: Professor Perry Samson, Professor Mark Flanner, and Professor Jeremy Bassis

- Visited Kangerlussuaq, Russell Glacier, and Summit Station in Greenland over two weeks
- Gained significant field research experience by working with team of students and professors to conduct multiple experiments involving stream flow, net solar radiation, PIBAL and rawinsonde launches, air quality, and 3D-scanning with drones
- Led and trained sub-team responsible for radiosonde launches

Research Assistant, University of Michigan 2018 – 2019
Project: **Ensemble Probability Program for Backward Trajectories from the University of Michigan Biological Station (UMBS) based on the Relative Abundance of δ^{18}O**
Advisor: Professor Perry Samson

- Wrote Python program that accessed a data bank of HY-SPLIT trajectories and retrieved trajectories corresponding to high or low δ^{18}O values, then ran trajectory coordinates through a probability distribution function to produce a map that described where air with a high or low δ^{18}O likely originated in order to find moisture sources for the UMBS using Python
- Concluded project by writing a summary paper and giving a poster presentation

Research Assistant, University of Michigan 2018 – 2019
Project: **Watersheds, δ^{18}O, and Climate of the Peruvian Andes**
Advisor: Dr. Phoebe Aron

- Interpolated meteorological, hydrological, and isotopic data with ArcGIS tools
- Developed and evaluated multiple procedures for calculating statistics of watersheds with ArcGIS tools
- Maintained a database of meteorological and isotopic information with SQL for Southern Peru
- Performed linear regression analysis on reanalysis datasets and isotope data using MATLAB
- Assisted PhD student relating to their study of hydrology in the Peruvian Andes by completing a variety of small research projects

Computer Skills

- Python – 8 years
- Jupyter – 5 years
- C++ – 5 years
- SQL – 2 years
- MATLAB – 4 years
- Fortran90 – 4 years
- Community Earth System Model (CESM) – 4 years
- ArcGIS – 4 years
- Unified Forecast System (UFS) – 1 year
- Autodesk Inventor – 2 years
- Adobe Illustrator – 1 years
- Microsoft PowerPoint – 4 years
- LaTeX – 7 years
- GitHub – 8 years
- High-Performance Computing (HPC) – 5 years
- Message-Passing Interface (MPI) – 1 year
- OpenMP – 1 year
- OpenACC – 1 year

Teaching Experience

Graduate Teaching Assistant, University of Colorado-Boulder — Fall 2026
Climate Modeling Lab (ATOC 4870/5870), Professor William Chapman
Planning to give a lecture in November 2026

Instructional Aide, University of Michigan — Fall 2021
Earth and Space System Evolution (CLIMATE 320), Professor Eric Kort

Collaborator, University of Michigan — Fall 2019 – Spring 2020
Melting Ice Rising Seas Teach-Out

Awards & Honors

ATOC Student Service Award, University of Colorado-Boulder — 2025
CIRES Graduate Student Research Award, University of Colorado-Boulder — 2025 – 2026
Best Poster in Aerosols, Clouds, and Precipitation, ESSS Conference, University of Colorado-Boulder — 2023
NSF Graduate Research Fellowship, University of Colorado-Boulder — 2022 – 2025
James B. Angell Scholar, University of Michigan — 2020 – 2022
Engineering Honors Program, University of Michigan — 2020 – 2022
William J. Branstrom Freshman Prize, University of Michigan — 2019
Dean's List, University of Michigan — 2018 – 2022

Service

Journal Reviewer
Geophysical Research Letters, Science Advances

ATOC RSO, University of Colorado-Boulder — 2024 – Present
- Helped organize and fund winter retreat and department SWAG

ATOC REU Mentor, University of Colorado-Boulder — 2024
- Mentored an undergraduate student for a summer research project
- Gave technical and career advice

RECCS Mentor, University of Colorado-Boulder — 2023
- Co-mentored an undergraduate student for a summer research project with another graduate student
- Gave technical and career advice

Prospective Student Committee, University of Colorado-Boulder — 2023 – Present
- Helped organized prospective student visits in the spring

Engineering Honors Graduate School Panel, University of Michigan — 2023 – 2025
- Served on five graduate school informational panels for seniors in the Engineering Honors program
- Answered questions about the graduate school application process and graduate school life

Graduate Application Mentorship Program Co-lead, University of Colorado-Boulder — 2022 – Present
- Mentored several students each year applying to the department by giving advice and reviewing personal statements
- Since 2023, co-led program recruiting mentors and matching mentors with mentees

Technology Committee, University of Colorado-Boulder 2022 – 2024

- Made improvements to department website including removal of old links and addition of photos and personal website links to graduate student pages
- Advocated for committee to have department website access to make regular updates

Executive Board Member for Out in STEM, University of Michigan 2020 – 2022

- Collaborated with other board members to plan a large event for welcome week at the University of Michigan
- Led and organized several weekly meetings and discussions
- Created online pen pal program to connect members with each other outside of weekly meetings to make up for the lack of in-person networking during COVID-19 pandemic

Languages

Latin (reading knowledge)

Professional Associations

American Geophysical Union
American Meteorological Society
Out in STEM

Publications

Submitted

Gilbert, A.L., Kay, J.E., and Blanchard-Wrigglesworth, E. Internal and anthropogenic contributions to recent (1950-2024) Arctic temperature and sea ice change. *Environmental Research: Climate.* Submitted.

In Review

Wright, D.M., Jablonowski, C., Fujisaki-Manome, A., Mroczka, B., **Gilbert, A.**, Titze, D., Mann, G.E., and Anderson, E.J. The Sensitivity of Lake-Effect Snowfall to Changes in Lake Surface Conditions Across the Forecast Horizon in the Unified Forecast System's Short-Range Weather Application (UFS-SRW). *Monthly Weather Review.* In review.

Published

Gale, S., Fu, Q., Wang, H., Blanchard-Wrigglesworth, E., Wang, M., **Gilbert, A.**, Sweeney, A.J., and Po-Chedley, S. (2026). Why April Stands Out: Monthly Impacts of Internal Variability on Arctic Amplification. *Journal of Geophysical Research: Atmospheres* **131**(19), e2026JD047413. [**PDF**]

Gilbert, A.L., Kay, J.E., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D.P. (2025). Observed winds alone cannot explain recent Arctic warming and sea ice loss. *Environmental Research: Climate* **4**(4), 045009. [**PDF**]

Gilbert, A., Kay, J.E., and Rowe, P. (2025). A novel model hierarchy isolates the limited effect of supercooled liquid cloud optics on infrared radiation. *Geoscientific Model Development* **18**(19), 7185–7197. [**PDF**]

Jonko, A., Oliveto, J., Beaty, T., Atchley, A., Battaglia, M.A., Dickinson, M.R., **Gilbert, A.**, Godwin, D., Kupfer, J.A., Hiers, J.K., Hoffman, C., North, M., Restaino, J., Sieg, C., and Skowronski, N. (2024). How will future climate change impact prescribed fire across the contiguous United States? *npj Climate and Atmospheric Science* **7**(1), 1–10. [**PDF**]

Snide, C.E., **Gilbert, L.**, Meyer, A., Samson, P., Flanner, M., and Bassis, J. (2020). Seeing the Greenland Ice Sheet through students' eyes. *Eos*, 101. [**PDF**]

Presentations

Seminars

Gilbert, A. Internal and anthropogenic contributions to recent (1950-2024) Arctic temperature sea ice change. ECS & Cloud Feedback Virtual Symposium. Virtual. September 2026.

Gilbert, A. Influence of winds and sea ice thickness on recent Arctic warming and sea ice loss. UCLouvain Earth and Life Institute: Climate. Louvain-la-Neuve, BE. June 2026.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Observed Winds Alone Cannot Explain Recent Arctic Warming and Sea Ice Loss. Atmospheric and Oceanic Sciences Colloquium. Boulder, CO. September 2025.

Talks

Gilbert, A., Kay, J., and Blanchard-Wrigglesworth, E. Attribution of Recent Arctic Warming and Sea Ice Loss in the Context of the Observed Winds. Community Earth System Model (CESM) Workshop. UCAR, Boulder, CO. June 2026.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Observed Winds Alone Cannot Explain Recent Arctic Warming and Sea Ice Loss. Climate and Cryosphere (CliC) Conference. Wellington, NZ. February 2026.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Community Earth System Model (CESM) Workshop. UCAR, Boulder, CO. June 2025.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. AMS Denver Summit 2025. Denver, CO. May 2025.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Schneider, D. What controls the rapid Arctic warming since 1980? International Conference on Arctic Research Planning (ICARP IV) at Arctic Science Summit Week 2025. Boulder, CO. March 2025.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Land Ice and Polar Climate Working Group Meeting 2025. NCAR, Boulder, CO. March 2025.

Gilbert, A., Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Polar Radiant Energy in the Far InfraRed Experiment (PREFIRE) Science Team Meeting. Boulder, CO. August 2024.

Gilbert, A., Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Community Earth System Model (CESM) Workshop. UCAR, Boulder, CO. June 2024.

Gilbert, A. and Kay, J. A Breakdown of Arctic Warming. Land Ice and Polar Climate Working Group Meeting 2024. NCAR, Boulder, CO. February 2024.

Posters

Gilbert, A., Kay, J., and Chepfer, H. An Observational Constraint on Future Greenland Precipitation Based on New Data from EarthCARE. AMS Madison Summit. Madison, WI. August 2026.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Observed Winds Alone Cannot Explain Recent Arctic Warming and Sea Ice Loss. Earth System and Space Science (ESSS) Conference. Boulder, CO. December 2025.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., and Schneider, D. What controls the rapid Arctic warming since 1980? American Geophysical Union (AGU) Abstracts. A33B-1975. Washington, DC. December 2024.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., and Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Earth System and Space Science (ESSS) Conference. Boulder, CO. December 2024.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., and Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Graduate Climate Conference (GCC). Pack Forest, Eatonville, WA. November 2024.

Gilbert, A., Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Cloud Feedback Modeling Intercomparison Project (CFMIP) Meeting. Boston College, Boston, MA. June 2024.

Gilbert, A., Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Polar Amplification of Climate Change Across Hemispheres and Seasons: Causes and Constraints Workshop. UCAR, Boulder, CO. January 2024.

Gilbert, A., Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Earth System and Space Science (ESSS) Conference. Boulder, CO. December 2023.

Gilbert, A. and Kay, J. Impact of Temperature Dependent Cloud Optical Properties on Modeled Infrared Radiation. Gordon Research Conferences (GRC) 2023 Radiation and Climate. Bates College, Lewiston, ME. July 2023.

Gilbert, L. and Bassis, J. Observing and Modeling Drainage Networks from Supraglacial Lakes on Russell Glacier, West Greenland. American Geophysical Union (AGU) Abstracts. C151C-0809. Virtual. December 2021.

Gilbert, L., Rowe, P., Fergoda, M., and Neshyba, S. Influence of Temperature Dependence of Cloud Optical Properties on Simulated IR Flux at South Pole, Antarctica. American Meteorological Society (AMS) Abstracts. 136. Virtual. January 2021.

Gilbert, L. and Bassis, J. Observing and Modeling Drainage Networks from Supraglacial Lakes on Russell Glacier, West Greenland. American Geophysical Union (AGU) Abstracts. C061-0011. Virtual. December 2020.

Gilbert, L., Aron, P., Samson, P., and Poulsen, C. Ensemble Probability Program for Backward Trajectories from the University of Michigan Biological Station (UMBS) based on the Relative Abundance of 18O. Michigan Geophysical Union (MGU) Symposium. Ann Arbor, MI. April 2019.